



RESEARCH

& INNOVATION UPDATES

Elevated 3D object tracking through invisible tagging

MIT scientists have created BrightMarkers, concealed fluorescent markers within 3D-printed objects like balls, containers, gadget cases, and gear, enhancing motion

tracking, virtual reality, and object detection. BrightMarkers are seamlessly integrated into 3D-printed objects using software plugins like Blender, saved as standard triangle

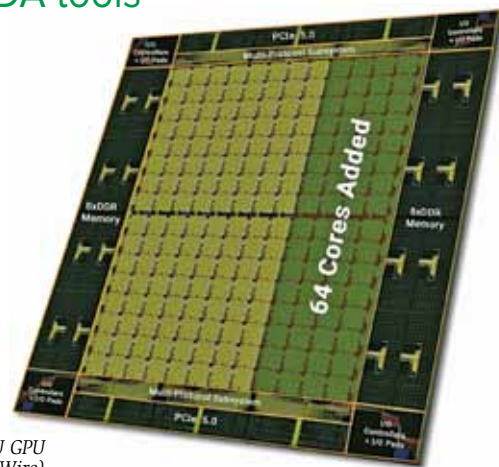


BrightMarker: 3D printed fluorescent markers for object tracking (Photos courtesy of the researchers/ MIT CSAIL)

language (STL) files for 3D printing, and printed with fluorescent filaments to form hidden, invisible QR code-like tags. These markers must be added during the fabrication process, offering specific near-infrared light emission detectable through infrared cameras, with two scanning setups designed for smartphones and AR/VR headsets. Applications range from supply chain management, motion tracking, and wearables for limb movement to night vision monitoring in home security cameras, preserving privacy. Limitations include the inability to add tags post-fabrication, potential obstructions like hands hindering tracking, and the potential for enhanced marker visibility with increased fluorochrome and magnetic filaments.

Revolutionising chip design with advanced EDA tools

Tachyum has announced that by employing new EDA tools during the physical design phase of their prodigy universal processor, using updated EDA tools, the team expanded the core count by 50% and increased SERDES to 96 per chip, slightly enlarging the die size to 600mm². They also enhanced the L2/L3 cache to 192MB, added DDR5 7200 memory support, and boosted the chip's speed with 1 DIMM per channel. This new design allows for an additional 32 serial links, facilitating connections to up to 32 DIMMs on a single prodigy chip. The patented prodigy architecture effortlessly toggles between standard CPU tasks and AI/ML duties, ideal for training and inference tasks, especially in the banking sector for improving financial security and identifying fraud proactively.



192-core prodigy processor Tachyum chip CPU TPU GPU AI supercomputing processor (Credit: Business Wire)